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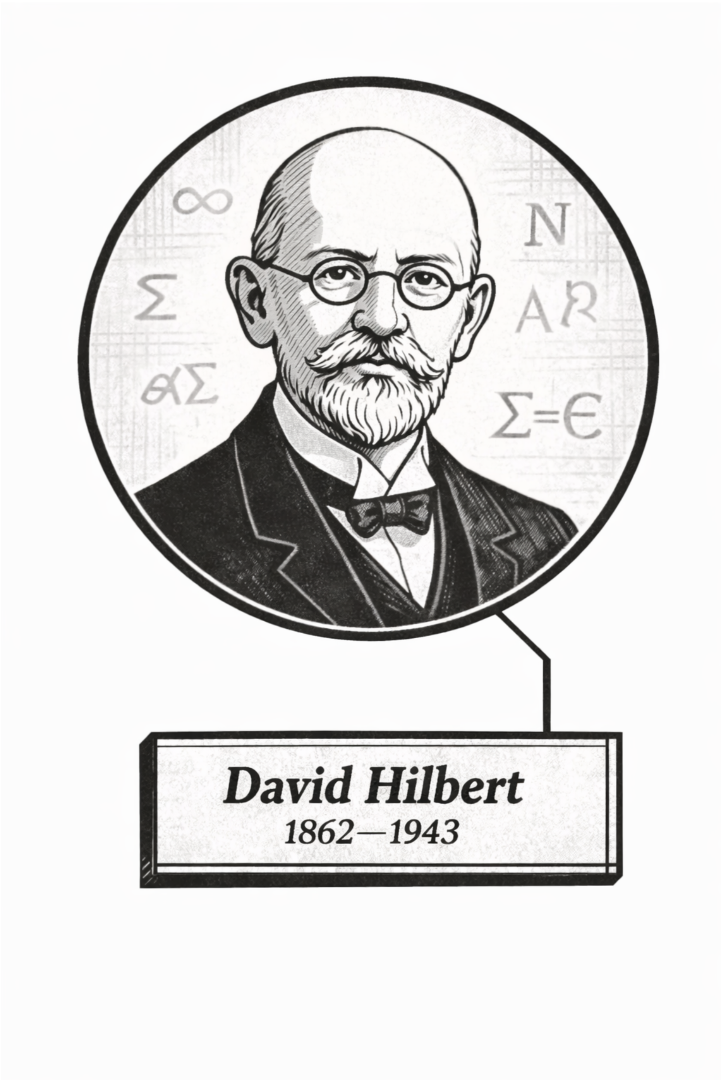

Frontispiece placeholder.
David Hilbert, 1862–1943

Volume VIII

Mathematical Logic and Foundations

Self-Study Notes

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June 9, 2026

Part I

Mathematical Logic and Foundations

Toolkit — Rings of Sets

Concept family: The ring axioms (Halmos–Givant ğ1) as the algebraic foundation for Boolean algebras of sets.

Atomic items in this section:

- Definition: Ring

Key predicate:

$$\mathrm{Ring}(R,+,\cdot,-,0,1) \;:\equiv\; \mathrm{AbelianGroup}(R,+,-,0) \wedge \mathrm{Monoid}(R,\cdot,1) \wedge \mathrm{Distributes}(\cdot,+,R).$$

Axiom numbering follows Halmos–Givant; axiom (4) (commutativity of multiplication) is not a ring axiom but a theorem in the Boolean setting.

Definition — Ring

Definition 0.0.1 (Ring). A **ring** is a non-empty set R together with two binary operations $+$ and $\cdot$ on R, a unary operation $-$ on R, and two distinguished elements $0, 1 \in R$, satisfying the following axioms for all $p, q, r \in R$:

(1) $p + (q + r) = (p + q) + r$ *(additive associativity)*

(2) $p \cdot (q \cdot r) = (p \cdot q) \cdot r$ *(multiplicative associativity)*

(3) $p + q = q + p$ *(additive commutativity)*

(5) $p + 0 = p$ *(additive identity)*

(6) $p \cdot 1 = p$ *(multiplicative identity)*

(7) $p + (-p) = 0$ *(additive inverse)*

(8) $p \cdot (q + r) = p \cdot q + p \cdot r$ *(left distributivity)*

(9) $(q + r) \cdot p = q \cdot p + r \cdot p$ *(right distributivity)*

Axiom numbering follows Halmos-Givant ?; axiom (4) is commutativity of multiplication, which is not a ring axiom but a theorem in the Boolean setting. A ring possessing a distinguished element 1 satisfying axiom (6) is called a **ring with unit**. A ring satisfying $p \cdot q = q \cdot p$ for all $p, q \in R$ is called a **commutative ring**.

Remark (Standard quantified statement).

$$
\begin{aligned}
(R,+,\cdot,-,0,1) \texttt{ is a ring} \iff \forall p,q,r \in R:& \\
& p+(q+r) = (p+q)+r \\
\wedge \quad & p\cdot(q\cdot r) = (p\cdot q)\cdot r \\
\wedge \quad & p+q = q+p \\
\wedge \quad & p+0 = p \\
\wedge \quad & p\cdot 1 = p \\
\wedge \quad & p+(-p) = 0 \\
\wedge \quad & p\cdot(q+r) = p\cdot q + p\cdot r \\
\wedge \quad & (q+r)\cdot p = q\cdot p + r\cdot p.
\end{aligned}
$$

Remark (Definition predicate reading).

$$\mathrm{Ring}(R,+,\cdot,-,0,1) \iff \mathrm{AbelianGroup}(R,+,-,0) \wedge \mathrm{Monoid}(R,\cdot,1) \wedge \mathrm{Distributes}(\cdot,+,R)$$

Remark (Negated quantified statement).

$$
\begin{aligned}
(R,+,\cdot,-,0,1) \texttt{ is } \textbf{not} \texttt{ a ring} \iff \exists p,q,r \in R:& \\
& p+(q+r) \neq (p+q)+r \\
\vee \quad & p\cdot(q\cdot r) \neq (p\cdot q)\cdot r \\
\vee \quad & p+q \neq q+p \\
\vee \quad & p+0 \neq p \\
\vee \quad & p\cdot 1 \neq p \\
\vee \quad & p+(-p) \neq 0 \\
\vee \quad & p\cdot(q+r) \neq p\cdot q + p\cdot r \\
\vee \quad & (q+r)\cdot p \neq q\cdot p + r\cdot p.
\end{aligned}
$$

Remark (Failure mode decomposition). The eight axioms partition into three structural groups:

$$\underbrace{(1),(3),(5),(7)}_{\mathrm{AbelianGroup}(R,+,-,0)} \qquad \underbrace{(2),(6)}_{\mathrm{Monoid}(R,\cdot,1)} \qquad \underbrace{(8),(9)}_{\mathrm{Distributes}(\cdot,+,R)}$$

A structure violating any axiom in the first group lacks a coherent additive group. A structure violating any axiom in the second group lacks a multiplicative identity or associativity. A structure violating the third group possesses two individually well-behaved operations that do not interact coherently; distributiv is the sole structural bridge between $+$ and $\cdot$, and its failure is sufficient to destroy the ring property.

Remark (Interpretation). A ring abstracts the arithmetic of the integers. The additive structure is always a commutative group. The multiplicative structure is a monoid -- associative with unit -- but need not be commutative; axiom (4) of Halmos-Givant, commutativity of multiplication, is absent from the ring axioms. In the Boolean setting, multiplicative commutativity is not assumed but derived as a theorem from idempotence alone. Distributivity encodes the coherent interaction between the two operations: it is the axiom that makes $\cdot$ and $+$ form a single algebraic system rather than two independent ones.

Toolkit — Boolean Rings

Concept family: Boolean rings (Halmos–Givant ğ1) and their immediate consequences from the single idempotence axiom.

Atomic items in this section:

- Definition: Boolean Ring
- Definition: Idempotent Element
- Definition: Characteristic 2
- Lemma: Self Additive Inverse ($p + p = 0$)
- Lemma: Self Negation ($-p = p$)

Key predicate:

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \;:\equiv\; \mathrm{Ring}(R, +, \cdot, -, 0, 1) \;\wedge\; \mathrm{IdempotentRing}(R, \cdot).$$

Central theorem chain: Idempotence $\Rightarrow$ Characteristic 2 $\Rightarrow$ Self Negation ($-p = p$) $\Rightarrow$ Commutativity of $\cdot$.

Definition — Boolean Ring

Definition 0.0.2 (Boolean Ring). A **Boolean ring** is an idempotent ring with unit.
That is, a Boolean ring is a ring $(R, +, \cdot, -, 0, 1)$ satisfying the additional axiom:

$$p \cdot p = p \quad \text{for all } p \in R. \tag{11}$$

An element p satisfying $p \cdot p = p$ is called **idempotent**. A ring in which every element is idempotent is called an **idempotent ring**.
The official axiom set for a Boolean ring consists of axioms (1)-(3), (5), (6), (8)-(11), where axiom (7) (the additive inverse law) is replaced by axiom (10) (characteristic 2), which is derived as a theorem. Axiom numbering follows Halmos-Givant ?.

Go to proof that every Boolean ring has characteristic 2.

Remark (Standard quantified statement).

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \mathrm{Ring}(R, +, \cdot, -, 0, 1) \wedge \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading).

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \mathrm{Ring}(R, +, \cdot, -, 0, 1) \wedge \mathrm{IdempotentRing}(R, \cdot)$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \neg\, \mathrm{Ring}(R, +, \cdot, -, 0, 1) \vee \exists p \in R\colon p \cdot p \neq p.$$

Remark (Failure mode decomposition). A structure fails to be a Boolean ring in exactly one of two ways:

$$\underbrace{\neg\ \mathrm{Ring}(R,+,\cdot,-,0,1)}_{\texttt{fails a ring axiom}} \qquad \vee \qquad \underbrace{\exists p \in R\colon p\cdot p \neq p}_{\texttt{some element is not idempotent}}$$

The second failure mode is the structurally interesting one. A ring in which even one element fails $p \cdot p = p$ is not Boolean. The simplest example: the ring $\mathbb{Z}$ of integers fails idempotence at every element except 0 and 1.

Remark (Canonical examples).

- $\mathbf{2} = \{0,1\}$ with addition mod 2 and ordinary multiplication is the simplest non-degenerate Boolean ring.
- $\mathbf{2}^n$, the set of all n-tuples of elements of $\mathbf{2}$, with coordinatewise addition and multiplication, is a Boolean ring with 2^n elements. The zero is $(0,\dots,0)$ and the unit is $(1,\dots,1)$.
- $(\mathcal{P}(X),\triangle,\cap)$, the power set of any set X with symmetric difference as addition and intersection as multiplication, is a Boolean ring. Under the correspondence $A \leftrightarrow \mathbf{1}_A$ (the characteristic function of A), this ring is isomorphic to $\mathbf{2}^X$.

Remark (Interpretation). A Boolean ring is the minimal algebraic structure that forces the full Boolean calculus to emerge from a single additional axiom -- idempotence of multiplication. The ring axioms supply the additive abelian group and the multiplicative monoid with distributivity. Idempotence then forces characteristic 2 (every element equals its own additive inverse), forces commutativity of multiplication, and identifies the ring with a field of sets via the representation theorem (Halmos-Givant, Chapter 22). The abstract definition thus captures exactly the algebra of set-theoretic operations: symmetric difference plays the role of addition and intersection plays the role of multiplication.

Remark (Dependencies).

- Definition: Ring

The simplest non-degenerate Boolean ring is the two-element ring $\mathbf{2} = \{0,1\}$. Its addition and multiplication are given by the following tables.

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Addition in $\mathbf{2}$ is addition modulo 2: $1+1=0$. Multiplication in $\mathbf{2}$ is ordinary multiplication of integers restricted to $\{0,1\}$. Every element of $\mathbf{2}$ satisfies $p+p=0$ and $p\cdot p = p$, making $\mathbf{2}$ the canonical example against which both axioms (10) and (11) are verified.

Definition — Idempotent Element

Definition 0.0.3 (Idempotent Element). An element p of a ring R is called **idempotent** if

$$p\cdot p = p.$$

A ring in which every element is idempotent is called an **idempotent ring**.

Remark (Standard quantified statement).

$$\mathrm{IdempotentElement}(p, R, \cdot) \iff p \cdot p = p.$$
$$\mathrm{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading).

$$\mathrm{IdempotentElement}(p, R, \cdot) \qquad \mathrm{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon \mathrm{IdempotentElement}(p, R, \cdot)$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{IdempotentElement}(p, R, \cdot) \iff p \cdot p \neq p.$$

Remark (Interpretation). Idempotence is the single additional axiom that forces Boolean structure. In the two-element ring $\mathbf{2}$, verification is immediate: $0 \cdot 0 = 0$ and $1 \cdot 1 = 1$. In a power set algebra $\mathcal{P}(X)$, idempotence of intersection -- $A \cap A = A$ -- is the set-theoretic counterpart. The force of the axiom is not in these examples but in what it implies for an arbitrary ring: characteristi and commutativity of multiplication both follow from idempotence alone, as the theorems below establish.

Remark (Dependencies).

- Definition: Ring

Definition — Characteristic 2

Definition 0.0.4 (Characteristic 2). A ring R is said to have **characteristic** 2 if

$$p + p = 0 \qquad \text{for all } p \in R. \tag{10}$$

Remark (Standard quantified statement).

$$\mathrm{Characteristic2}(R, +, 0) \iff \forall p \in R\colon p + p = 0.$$

Remark (Definition predicate reading).

$$\mathrm{Characteristic2}(R, +, 0) \iff \forall p \in R\colon \mathrm{IdempotentElement}(p, R, +)$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{Characteristic2}(R, +, 0) \iff \exists p \in R\colon p + p \neq 0.$$

Remark (Interpretation). Characteristic 2 means that every element is its own additive inverse: $p+p = 0$ says precisely $p = -p$. The operation of negation therefore becomes the identity map -- applying $-$ to any element returns that element unchanged. Halmos observes that in a ring of characteristic 2, the minus sign for additive inverses is never needed and is henceforth dropped. Axiom (7), $p+(-p) = 0$, is replaced in the official Boolean ring axiom set by the stronger equation (10), $p + p = 0$, which is a theorem derived from idempotence.

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Idempotent Element

Lemma 0.0.5 (Self Additive Inverse). *In any Boolean ring R,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R \colon p + p = 0.$$

Remark (Definition predicate reading).

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \implies \mathrm{Characteristic2}(R, +, 0).$$

Remark (Interpretation). This is Halmos consequence (a): idempotence of multiplicatio forces the additive structure to have characteristic 2. The proof expands $(p+q)^2$ by distributivity and idempotence, cancels p and q, then sets $p = q$ to obtain $p+p = 0$. The Lean 4 formalisation is AdditiveIdempotence in BooleanAlgebra

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Characteristic 2

Lemma 0.0.6 (Self Negation). *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R \colon -p = p.$$

Remark (Interpretation). This follows immediately from 0.0.5. Since $p+p = 0$ and $p+(-p) = 0$ both hold, the uniqueness of additive inverses gives $-p = p$. The negation operation is therefore the identity map on any Boolean ring. Halmos uses this to drop the minus sign entirely from his notation for the rest of the book: since $-p = p$, writing $-p$ and writing p are the same thing.

Remark (Dependencies).

- Lemma: Self Additive Inverse

Toolkit — Boolean Ring Examples

Concept family: Concrete Boolean ring examples and structural constraints.

Atomic items in this section:

- Remark: No Boolean ring of order 3 exists
- Example: The four-element Boolean ring $\mathbf{2}^2$
- Remark: Boolean rings have order 2^n

Key constraint: $(R, +)$ in any Boolean ring is an elementary abelian 2-group, so $|R| = 2^n$ for some $n \geq 0$.

Remark (No Boolean ring of order 3). A Boolean ring of order 3 does not exist. The additive group $(R,+)$ of any Boolean ring satisfies $p+p=0$ for all p, by Lemma 0.0.5. Every element therefore has additive order dividing 2, so $(R,+)$ is an elementary abelian 2-group. Elementary abelian 2-groups have order 2^n for some $n \geq 0$. Since 3 is not a power of 2, no Boolean ring of order 3 can exist.

Example (The four-element Boolean ring $\mathbf{2}^2$). The smallest Boolean ring with more than two elements is $\mathbf{2}^2 = \{(0,0), (0,1), (1,0), (1,1)\}$, the set of ordered pairs from $\mathbf{2} = \{0,1\}$. Addition and multiplication are defined coordinatewise:

$$(p_0,p_1)+(q_0,q_1) \;=\; (p_0+q_0,\; p_1+q_1), \qquad (p_0,p_1)\cdot(q_0,q_1) \;=\; (p_0\cdot q_0,\; p_1\cdot q_1),$$

where each coordinate is computed in $\mathbf{2}$. The zero element is $(0,0)$ and the unit is $(1,1)$.

Addition table.

$+$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,1)$	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
$(1,0)$	$(1,0)$	$(1,1)$	$(0,0)$	$(0,1)$
$(1,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$

Multiplication table.

$\cdot$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$	$(0,1)$	$(0,0)$	$(0,1)$
$(1,0)$	$(0,0)$	$(0,0)$	$(1,0)$	$(1,0)$
$(1,1)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$

Verification of idempotence. Every element squares to itself:

p	$p\cdot p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,1)$
$(1,0)$	$(1,0)$
$(1,1)$	$(1,1)$

Verification of characteristic 2. Every element is its own additive inverse:

p	$p+p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$
$(1,0)$	$(0,0)$
$(1,1)$	$(0,0)$

Remark (Boolean rings have order 2^n). Every finite Boolean ring is isomorphic to $\mathbf{2}^n$ for some $n \geq 0$, and therefore has exactly 2^n elements. This follows from the representation theorem (Halmos-Givant, Chapter 22): every Boolean algebra is isomorphic to a field of sets, and every finite field of sets is isomorphic to $\mathcal{P}(X)$ for some finite set X with $|X| = n$, giving $|\mathcal{P}(X)| = 2^n$. The permissible finite sizes are $1, 2, 4, 8, 16, \ldots$ and no other cardinality is achievable.

Capstone: The Power Set Boolean Ring

Remark (Return). Chapter capstone -- synthesises Sections 2, 3, and 4.

Remark (Capstone Problem). Let X be a set. Prove that $(\mathcal{P}(X), \triangle, \cap)$ is a Boolean ring, where $\triangle$ denotes symmetric difference and $\cap$ denotes intersection.

Specifically, verify:

(i) $(\mathcal{P}(X), \triangle)$ is an abelian group with identity $\emptyset$.

(ii) $(\mathcal{P}(X), \cap)$ is a commutative monoid with identity X.

(iii) $\cap$ distributes over $\triangle$.

(iv) Every element is idempotent under $\cap$: $A \cap A = A$.

Do not re-prove the named propositions; assemble them.

Proof. **Professional Standard Proof.**

We verify each condition in turn, assembling the named propositions referenced in the Dependencies block.

(i) $(\mathcal{P}(X), \triangle)$ is an abelian group with identity $\emptyset$.

- *Closure.* For any $A, B \in \mathcal{P}(X)$, $A \triangle B = (A \setminus B) \cup (B \setminus A) \subseteq X$, so $A \triangle B \in \mathcal{P}(X)$.
- *Commutativity.* By prop:symmdiff-comm: $A \triangle B = B \triangle A$.
- *Associativity.* By prop:symmdiff-assoc: $(A \triangle B) \triangle C = A \triangle (B \triangle C)$.
- *Identity.* By prop:symmdiff-empty: $A \triangle \emptyset = A$. So $\emptyset$ is the additive identity.
- *Inverse.* By prop:symmdiff-self: $A \triangle A = \emptyset$. So every element is its own inverse, confirming characteristic 2 directly.

Therefore $(\mathcal{P}(X), \triangle, \emptyset)$ is an abelian group.

(ii) $(\mathcal{P}(X), \cap)$ is a commutative monoid with identity X.

- *Closure.* For $A, B \in \mathcal{P}(X)$, $A \cap B \subseteq A \subseteq X$, so $A \cap B \in \mathcal{P}(X)$.
- *Associativity.* By prop:assoc-intersection: $(A \cap B) \cap C = A \cap (B \cap C)$.
- *Commutativity.* By prop:comm-intersection: $A \cap B = B \cap A$.
- *Identity.* By prop:identity-intersection: $A \cap X = A$. So X is the multiplicative identity.

Therefore $(\mathcal{P}(X), \cap, X)$ is a commutative monoid.

(iii) $\cap$ distributes over $\triangle$.

By prop:cap-dist-symmdiff: for all $A, B, C \in \mathcal{P}(X)$,

$$A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C).$$

Since $\cap$ is commutative, right distributivity follows: $(B\triangle C)\cap A = A\cap(B\triangle C) = (A\cap B)\triangle(A\cap C) = (B\cap A)\triangle(C\cap A)$. Both distributive laws hold.

(iv) Every element is idempotent under $\cap$.
By prop:idempotence-intersection: $A\cap A = A$ for every $A \in \mathcal{P}(X)$.

Conclusion. Parts (i)-(iii) establish the ring axioms (1)-(3),(5)-(9) of Halmos-Givant. Part (iv) is axiom (11) (multiplicative idempotence). The ring is commutative by prop:comm-intersection (already used in (ii)), and X serves as the multiplicative unit. Therefore $(\mathcal{P}(X), \triangle, \cap)$ is a commutative Boolean ring with unit. ☐

Remark (Proof shape). The proof is a direct assembly. Each of the four conditions maps onto a named proposition in the notes. The key insight is the correspondence:

$$\text{symmetric difference} \longleftrightarrow \text{addition}, \qquad \text{intersection} \longleftrightarrow \text{multiplication},$$

$$\emptyset \longleftrightarrow 0, \qquad X \longleftrightarrow 1.$$

The characteristic 2 property ($A\triangle A = \emptyset$, i.e., every element is its own additive inverse) means the ring axiom (7) follows from axiom (11) (idempotence), exactly as the Self Additive Inverse lemma shows in the abstract setting. The power set ring is therefore the canonical model that makes the Boolean ring abstraction concrete.

Connection to simulation. In the Brownian motion simulation on the torus, the event algebra tracking which particles have met (or are within a threshold distance) forms precisely such a Boolean ring of subsets under symmetric difference and intersection. The characteristic 2 property has a concrete meaning: asking whether two events $A\triangle A$ have occurred reduces immediately to $\emptyset$ -- the vacuous event. The ring structure is what makes the event-algebra queries compositionally well-typed.

Remark (Dependencies).

- Definition: Ring
- Definition: Boolean Ring
- Definition: Idempotent Element

Remark (Return). Return to Definition: Ring

Remark (Note on proof type). This file accompanies the Ring definition rather than a theorem. The "proof" here is a verification that the axiom set is internally consistent (no axiom is derivable from the others in a way that would create circular dependencies) and that the structure as given is non-empty by example. It is not a proof of a theorem but a justification that the definition is well-posed.

Proposition (The Ring Axioms Are Well-Defined). *The axiom set {(1),(2),(3),(5),(6),(* *of Halmos-Givant is non-redundant and consistent: each axiom fails in some structure satisfying all the others, and the integers* $(\mathbb{Z}, +, \cdot, -, 0, 1)$ *witness consistency.*

Proof. TODO ∎

Remark (Return). Return to Definition: Boolean Ring

Remark (Relationship to existing proof). The proof of this theorem is contained in full in Self Additive Inverse (prf:boolean-ring-self-additive-inverse), which proves $p+p=0$ directly from idempotence. The label prf:boolean-ring-characteristic-2 is the entry point linked from the Boolean Ring definition; the content is shared.

Theorem (Every Boolean Ring Has Characteristic 2). *In any Boolean ring R,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Equivalently, $\mathrm{Characteristic2}(R, +, 0)$ *holds in every Boolean ring.*

Proof. **Professional Standard Proof.**
Apply idempotence to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

Expand the left side using right distributivity (9), then left distributivity (8), then collapse each $p \cdot p = p$ by idempotence (11):

$$(p+p)\cdot(p+p) \overset{(9)}{=} p\cdot(p+p)+p\cdot(p+p) \overset{(8)}{=} (p^2+p^2)+(p^2+p^2) \overset{p^2=p}{=} (p+p)+(p+p). \tag{ii}$$

Combining (i) and (ii): $(p + p) + (p + p) = p + p$. Setting $s = p + p$, the equation $s + s = s$ gives $s = 0$ by adding $-s$ and applying associativity, the inverse law, and the identity law. Hence $p + p = 0$. ☐

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Idempotent Element

Remark (Return). Return to Lemma

Lemma (Self Additive Inverse). *In any Boolean ring* R*,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Proof. **Professional Standard Proof.**
Apply idempotence to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

Expand the left side using right distributivity (9), then left distributivity (8), then collapse each $p \cdot p = p$ by idempotence (11):

$$(p+p)\cdot(p+p) \overset{(9)}{=} p\cdot(p+p)+p\cdot(p+p) \overset{(8)}{=} (p^2+p^2)+(p^2+p^2) \overset{p^2=p}{=} (p+p)+(p+p). \tag{ii}$$

Combining (i) and (ii): $(p + p) + (p + p) = p + p$. Setting $s = p + p$, the equation $s + s = s$ gives $s = 0$ by adding $-s$ and applying associativity, the inverse law, and the identity law. Hence $p + p = 0$. □

Proof. **Detailed Learning Proof.**

Step 1. Apply idempotence to the element $p+p$**.** The element $p+p$ is a member of the carrier of R. Axiom (11) (multiplicative idempotence) asserts every element of R is its own square. Apply this to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

This is the pivot of the proof. It says $p + p$ squares to itself. Nothing about $p + p = 0$ has been assumed; that is the conclusion to be derived.

Step 2. Expand $(p+p)\cdot(p+p)$ **using right distributivity.** Axiom (9) (right distributivity) states:

$$(q + r) \cdot t = q \cdot t + r \cdot t \qquad \forall\, q, r, t \in R.$$

Set $q = p$, $r = p$, $t = p + p$:

$$(p + p) \cdot (p + p) = p \cdot (p + p) \;+\; p \cdot (p + p). \tag{ii}$$

The product has split into two identical copies of $p\cdot(p+p)$. This step holds in any ring; no idempotence has been used.

Step 3. Expand each $p\cdot(p+p)$ **using left distributivity.** Axiom (8) (left distributivity) states:

$$t \cdot (q + r) = t \cdot q + t \cdot r \qquad \forall\, t, q, r \in R.$$

Apply to $p \cdot (p + p)$ with $t = p$, $q = p$, $r = p$:

$$p \cdot (p + p) = p \cdot p \;+\; p \cdot p.$$

Apply to both copies in (ii):

$$p \cdot (p + p) + p \cdot (p + p) = (p \cdot p + p \cdot p) + (p \cdot p + p \cdot p). \tag{iii}$$

The right side contains four occurrences of $p \cdot p$. This expansion holds in any ring; idempotence has still not been used.

Step 4. Collapse each $p{\cdot}p$ to p by idempotence. Axiom (11) gives $p{\cdot}p = p$. Substitute into all four occurrences in (iii):

$$(p \cdot p + p \cdot p) + (p \cdot p + p \cdot p) = (p + p) + (p + p). \tag{iv}$$

This is the second and final use of idempotence.

Step 5. Chain (i) through (iv). Equations (i), (ii), (iii), (iv) combine as:

$$p + p \;=\; (p + p) \cdot (p + p) \;=\; (p + p) + (p + p).$$

Hence

$$(p + p) + (p + p) = p + p. \tag{v}$$

Step 6. Conclude $p{+}p = 0$ by cancellation. Let $s = p{+}p$. Equation (v) states $s + s = s$. Add the additive inverse $-s$ to the right of both sides, then apply associativity, the inverse law, and the identity law:

$$s{+}s = s \implies (s{+}s){+}(-s) = s{+}(-s) \implies s{+}\underbrace{(s + (-s))}_{=0} = 0 \implies s{+}0 = 0 \implies s = 0.$$

Since $s = p + p$, the conclusion is $p + p = 0$. $\square$

Remark (Proof structure). The proof is direct. The strategy applies idempotence twice: once to $p{+}p$ as a whole (Step 1) and once to p individually (Step 4), with distributivity (Steps 2-3) connecting the two applications. The conclusion follows by a standard additive cancellation (Step 6) using only the group axioms for $(R, +)$. No circularity arises: the proof never assumes $1{+}1 = 0$ or any consequence of characteristic 2. Both $p{+}p = 0$ and $1{+}1 = 0$ are consequences of idempotence; this proof derives the former directly without appealing to the latter. The Lean 4 formalisation is AdditiveIdempotence in LRA/VolumeI/BooleanAl

Remark (Dependencies).

- Definition of Ring -- axioms (1) associativity, (5) identity, (7) inverse, (8) left distributivity, (9) right distributivity.
- Definition of Boolean Ring -- axiom (11) multiplicative idempotence, applied in Steps 1 and 4.
- Definition of Idempotent Element -- names the property $p \cdot p = p$.

Remark (Return). Return to Lemma

Lemma (Self Negation). *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Proof. **Professional Standard Proof.**
By Self Additive Inverse, $p+p=0$. By the additive inverse axiom (7), $p+(-p)=0$. Since both p and $-p$ satisfy the equation $p+x=0$, and additive inverses are unique in any group (a standard group theory fact applied to the abelian group $(R,+)$), we conclude $-p=p$. □

Proof. **Detailed Learning Proof.**

Step 1. Recall the two equations that both hold in R.
From Self Additive Inverse (Lemma 0.0.5):

$$p + p = 0. \tag{i}$$

From the additive inverse axiom (7) of the ring definition:

$$p + (-p) = 0. \tag{ii}$$

Step 2. Apply uniqueness of additive inverses.
The additive group $(R,+,0)$ is a group (abelian, by axiom (3)). In any group, if $a+b=0$ and $a+c=0$, then:

$$b = 0 + b = (a+c) + b \overset{\text{assoc}}{=} a + (c+b) \overset{\text{comm}}{=} a + (b+c) \overset{\text{assoc}}{=} (a+b) + c = 0 + c = c.$$

Apply with $a=p$, $b=p$ (from (i)) and $c=-p$ (from (ii)): both p and $-p$ are additive inverses of p, so $p=-p$. □

Remark (Proof structure). A one-step uniqueness argument. Self Additive Inverse shows p is its own additive inverse; the ring axiom asserts $-p$ is also an additive inverse of p; uniqueness of inverses in the underlying group forces $p=-p$. The proof has no calculation content -- it is purely structural.

Remark (Dependencies).

- Lemma: Self Additive Inverse
- Definition: Ring (axiom (7), additive inverse law)

Chapter 1

Category Theory

Status: Planned

Active mathematical content has not yet been added.

Category Theory Notes

This chapter is planned. Active mathematical content has not yet been added.

Proofs

Proof entries for this chapter are planned. No proof files have been regenerated.

Exercises

Exercises for this chapter are planned. No exercise content has been restored here yet.

Chapter 2

Model Theory

Where You Are in the Journey

Proof Techniques $\rightarrow$ **Model Theory** $\rightarrow$ Type Theory $\rightarrow$ $\cdots$

How we got here. The preceding chapters established the syntax and proof theory of first-order logic: well-formed formulas, deduction rules, and proof techniques. Predicate logic gave a language; proof techniques gave a calculus. What has not yet been addressed is *meaning* the assignment of mathematical content to logical symbols.

What this chapter builds. Model theory is the study of the relationship between formal languages and the mathematical structures that interpret them. The central objects are $\mathcal{L}$-structures: a domain equipped with interpretations of every function symbol, relation symbol, and constant in a first-order language $\mathcal{L}$. From structures, the chapter develops the satisfaction relation $\mathcal{M} \models \varphi[s]$, which gives formal meaning to truth-in-a-structure. Elementary equivalence, substructures, and the compactness theorem are developed in later sections.

Where this leads. Model theory supplies the semantic foundations for abstract algebra (a group is an $\mathcal{L}_{\text{gp}}$-structure satisfying the group axioms), for real analysis (the ordered field axioms characterise $\mathbb{R}$ up to isomorphism among Archimedean fields), and ultimately for the ultraproduct constructions that appear in non-standard analysis and the compactness arguments used throughout the curriculum.

Structural Roadmap

Each major topic is organised as:

Definitions $\longrightarrow$ **Main Theorems** $\longrightarrow$ **Consequences and Structural Insight**

The global progression is:

1. **First-order languages and $\mathcal{L}$-structures.** Syntax of languages; domains, function interpretations, relation interpretations, constant interpretations.

2. **Variable assignments and term evaluation.** How terms acquire values in a structure.

3. **The satisfaction relation.** Inductive definition of $\mathcal{M} \models \varphi[s]$; truth and falsity in a structure.

4. **Elementary equivalence and substructures.** $\mathcal{M} \equiv \mathcal{N}$, $\mathcal{N} \prec \mathcal{M}$, the Tarski–Vaught criterion.

5. **Theories and models.** *[Planned]*

6. **Compactness and Löwenheim–Skolem.** *[Planned]*

Remark 2.0.1 (Primary sources). Marker, *Model Theory: An Introduction* (Springer GTM 217), Chapter 1. Enderton, *A Mathematical Introduction to Logic*, Chapter 2. Hodges, *A Shorter Model Theory*, Chapter 1.

2.1 Languages and Structures

L-Structures Toolkit Quick Reference

Concept	Symbol / Notation	Detail
First-order language	$\mathcal{L}$	
L-structure	$\mathcal{M} = (M, \mathcal{L}^{\mathcal{M}})$	
Domain / universe	$\lvert\mathcal{M}\rvert$ or M	
Interpretation of f	$f^{\mathcal{M}} : M^n \to M$	
Interpretation of R	$R^{\mathcal{M}} \subseteq M^n$	
Interpretation of c	$c^{\mathcal{M}} \in M$	
Variable assignment	$s : \mathrm{Var} \to M$	
Term evaluation	$s(t) \in M$	
Satisfaction	$\mathcal{M} \models \varphi[s]$	

2.2 First-Order Languages

Definition (First-Order Language $\mathcal{L}$)

A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \to, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

Remark (Fully quantified form). $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ where $\mathrm{ar}(f) \in \mathbb{N}$ for each $f \in \mathcal{F}$ and $\mathrm{ar}(R) \in \mathbb{N}_{\geq 1}$ for each $R \in \mathcal{R}$.

Remark (Intuition). A language is a *vocabulary*: it names the operations, relations, and distinguished elements that can appear in sentences, without yet saying what those names mean. The meaning is supplied by a structure.

Remark (Source note). Some sources (e.g. Marker §1.1, Hodges) absorb constant symbols into 0-ary function symbols. Others (e.g. Chang-Keisler) list them separately. The content is equivalent; only the bookkeeping differs.

Remark (Consequence). The set of $\mathcal{L}$-terms and the set of $\mathcal{L}$-formulas are both defined inductively from $\mathcal{L}$; see term evaluation below.

Example 2.2.1 (Standard examples of first-order languages).

1. **Language of ordered fields** $\mathcal{L}_{\mathrm{of}}$: function symbols $\{+, \cdot, -\}$ of arities $2, 2, 1$; relation symbol $\{<\}$ of arity 2; constant symbols $\{0, 1\}$. The real field $(\mathbb{R}, +, \cdot, -, <, 0, 1)$ is an $\mathcal{L}_{\mathrm{of}}$-structure.

2. **Language of graphs** $\mathcal{L}_{\mathrm{gr}}$: no function symbols, no constants; one binary relation symbol $\{E\}$ (for "edge").

3. **Language of groups** $\mathcal{L}_{\mathrm{gp}}$: function symbols $\{\cdot, {}^{-1}\}$ of arities $2, 1$; constant symbol $\{e\}$.

4. **Pure equality language** $\mathcal{L}_{=}$: no non-logical symbols at all. Structures are arbitrary non-empty sets.

2.2.1 L-Structures

Definition ($\mathcal{L}$-Structure)

Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$-structure** $\mathcal{M}$ consists of:

1. a non-empty set M called the **domain** (or **universe**) of $\mathcal{M}$, written $|\mathcal{M}|$;
2. for each n-ary relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq M^n$;
3. for each n-ary function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : M^n \to M$;
4. for each constant symbol $c \in \mathcal{C}$, a distinguished element $c^{\mathcal{M}} \in M$.

The superscript notation $f^{\mathcal{M}}, R^{\mathcal{M}}, c^{\mathcal{M}}$ denotes the **interpretation** of each symbol in $\mathcal{M}$.

Remark (Fully quantified form). $\mathcal{M}$ is an $\mathcal{L}$-structure iff $M \neq \emptyset$, and $\forall f \in \mathcal{F}_{(n)},\ f^{\mathcal{M}} : M^n \to M$, and $\forall R \in \mathcal{R}_{(n)},\ R^{\mathcal{M}} \subseteq M^n$, and $\forall c \in \mathcal{C},\ c^{\mathcal{M}} \in M$.

Remark (Intuition). A structure is an $\mathcal{L}$-vocabulary *made concrete*: each symbol is assigned an actual mathematical object of the correct type. The domain provides the raw material; the interpretations give the symbols their meaning inside that domain.

Remark (Common error). The domain M must be *non-empty*. First-order logic with an empty domain is pathological: $\forall x\, \varphi$ would be vacuously true and $\exists x\, \varphi$ vacuously false for every φ, breaking the duality of the quantifiers.

Remark (Source note). Some sources write $\mathfrak{M}$ (Fraktur) for structures and reserve M for the domain; others use $\mathcal{M}$ throughout and write $|\mathcal{M}|$ for the domain. The latter convention is used here. Marker uses $\mathcal{M}$; Enderton uses $\mathfrak{A}$.

Example 2.2.2 (L-structures for familiar mathematical objects). 1. $(\mathbb{R}, +^{\mathbb{R}}, \cdot^{\mathbb{R}}, -^{\mathbb{R}}, <^{\mathbb{R}}, 0^{\mathbb{R}}, 1^{\mathbb{R}})$ is an $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{R}$.

2. $(\mathbb{Q}, +^{\mathbb{Q}}, \cdot^{\mathbb{Q}}, -^{\mathbb{Q}}, <^{\mathbb{Q}}, 0^{\mathbb{Q}}, 1^{\mathbb{Q}})$ is another $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{Q}$.

3. The complete graph K_3 on $\{1, 2, 3\}$ is an $\mathcal{L}_{\text{gr}}$-structure with $E^{K_3} = \{(1, 2), (2, 1), (1, 3), ($

Two different structures can share the same language the language only fixes the *type* of the interpretations, not their specific content.

2.2.2 Variable Assignments

Definition (Variable Assignment)

Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain M, and let $\mathrm{Var} = \{v_1, v_2, v_3, \ldots\}$ be a fixed countable set of first-order variables.
A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \mathrm{Var} \to M$.
For a variable assignment s, a variable v_i, and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i, where it returns a:

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s[v_i \mapsto a](v_j) = a$ if $j = i$, else $s(v_j)$.

Remark (Intuition). A variable assignment gives temporary values to free variables before evaluating a formula. The $s[v_i \mapsto a]$ notation is the standard tool for handling quantifier rules: to check $\exists v_i\, \varphi$, test whether $\mathcal{M} \models \varphi[s[v_i \mapsto a]]$ for some $a \in M$.

2.2.3 Term Evaluation

Definition (Term Evaluation)

Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $s : \mathrm{Var} \to M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$-term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \ldots, t_n)$ for an n-ary function symbol f, then $s(t) = f^{\mathcal{M}}(s(t_1), \ldots, s(t_n))$.

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s(t) \in M$ for every $\mathcal{L}$-term t. Formally: $s(\cdot)$ is the unique function on terms satisfying clauses (1)-(3).

Remark (Intuition). Term evaluation is the "computation" step: given concrete values for all variables, every term reduces to an element of M. The inductive definition mirrors the inductive definition of terms themselves.

2.2.4 Satisfaction [Stub]

Definition (Satisfaction, $\mathcal{M} \models \varphi[s]$) Stub

Let $\mathcal{M}$ be an $\mathcal{L}$-structure, s a variable assignment, and φ an $\mathcal{L}$-formula. The **satisfaction relation** $\mathcal{M} \models \varphi[s]$ (read: "$\mathcal{M}$ satisfies φ under s") is defined inductively on the structure of φ:

[To be filled in clauses for atomic formulas, negation, conjunction, disjunction, implication, existential quantifier, universal quantifier.]

Remark (Fully quantified form). *[Stub to be completed.]*

Remark (Intuition). Satisfaction is the semantic notion of truth: $\mathcal{M} \models \varphi[s]$ means φ is true in $\mathcal{M}$ when each free variable v_i takes the value $s(v_i)$. The definition proceeds by induction on formula complexity, grounding out on atomic formulas evaluated via term evaluation.

2.2.5 Elementary Equivalence and Substructures [Stub]

Status: Planned

The following topics are planned for this section:

1. Substructures ($\mathcal{N} \subseteq \mathcal{M}$).
2. Elementary substructures ($\mathcal{N} \prec \mathcal{M}$) and the Tarski–Vaught criterion.
3. Elementary equivalence ($\mathcal{M} \equiv \mathcal{N}$).
4. Isomorphisms of $\mathcal{L}$-structures.

Chapter 3

Type Theory

Where You Are in the Journey

Propositional Logic $\rightarrow$ Predicate Calculus $\rightarrow$ Sets & Functions $\rightarrow$ Proof Techniques $\rightarrow$ Model Theory $\rightarrow$ **Type Theory** $\rightarrow \cdots$

How we got here. Set theory provides one foundation for mathematics. Type theory provides an alternative, assigning every expression a *type* to avoid paradoxes and to make proof-checking mechanical.

What this chapter will build. Simply typed lambda calculus, the Curry–Howard correspondence (proofs as programs), and dependent type theory as a foundation for mechanised proof assistants (Lean, Coq).

Where this leads. Modern proof assistants (Lean, Coq, Agda) are built on type theory. Understanding type theory illuminates what it means to formalise mathematics completely.

Status: Planned

Coming Soon

Notes, proofs, and exercises will appear here in a future revision.

To be populated.

Proofs

To be populated.

Capstone

To be populated.

Chapter 4

Lambda Calculus

Where You Are in the Journey

Propositional Logic $\to$ Predicate Calculus $\to$ Sets & Functions $\to$ Proof Techniques $\to$ Model Theory $\to$ Type Theory $\to$ **Lambda Calculus** $\leftarrow$ you are here

Chapter Overview

Lambda calculus (λ-calculus) is a formal system for expressing computation through function abstraction and application. Introduced by Alonzo Church (1936) to formalise computability, it is the theoretical foundation of functional programming and is equivalent in power to Turing machines.

Why it belongs here: Lambda calculus sits at the intersection of logic, set theory, and computation. The Curry–Howard correspondence connects it directly to the predicate logic and type theory in this volume: propositions are types, proofs are programs, and implication corresponds to function type.

Primary source: Barendregt & Barendsen, *Introduction to Lambda Calculus* (free PDF). Secondary: Rojas, *A Tutorial Introduction to the Lambda Calculus*. Bridge to types: Pierce, *Types and Programming Languages* Ch. 5, 8, 9.

Planned Content

1. **Syntax** — terms, variables, abstraction $\lambda x.M$, application MN, free and bound variables
2. **Reduction** — α-conversion, β-reduction, η-reduction, normal forms
3. **Church Encodings** — booleans, natural numbers (Church numerals), pairs, lists
4. **Fixed Points** — the Y-combinator, recursive functions
5. **Computability** — λ-definability, equivalence with Turing machines, undecidability of β-equality
6. **Typed Lambda Calculus** — simple types, type inference, Curry–Howard correspondence

Section Toolkit — Quick Reference

Concept	One-line summary
λ-term	Variable x, abstraction $\lambda x.M$, application MN
α-equivalence	$\lambda x.M =_\alpha \lambda y.M[y/x]$ when y fresh
β-reduction	$(\lambda x.M)N \to_\beta M[N/x]$
Normal form	Term with no β-redex; may not exist (non-termination)
Church boolean **T**	$\lambda x.\lambda y.x$
Church boolean **F**	$\lambda x.\lambda y.y$
Church numeral $\overline{n}$	$\lambda f.\lambda x.f^n(x)$ (f applied n times to x)
Type $\tau \to \sigma$	Function type: input of type τ, output of type σ
Typing judgment	$\Gamma \vdash M : \tau$ reads "M has type τ in context Γ"
Curry–Howard	Propositions are types; proofs are terms; $A \to B$ is implication

Untyped Lambda Calculus

Background References

- **First-order logic:** Enderton [1972], Barwise [1977a].
- **General topology:** Kelly [1955].
- **Set theory:** Halmos [1960], van Dalen et al. [1978].
- **Recursion theory:** Rogers [1967].
- **Category theory:** Arbib and Manes [1975], MacLane [1972].

Definition — Lambda Terms

Definition 4.0.1 (Lambda Terms). The set Λ of λ-terms is defined inductively as follows:

1. Every variable x is a λ-term.
2. If M is a λ-term and x is a variable, then
$$\lambda x.\, M$$
is a λ-term.
3. If M and N are λ-terms, then
$$MN$$
is a λ-term.

Remark (Interpretation). The λ-terms are the smallest set closed under three constructions: variables, abstraction (function formation), and application (function evaluation). This inductive definition plays the same role as recursive definitions in analysis or algebra: every λ-term is built from variables by finitely many applications of abstraction and application.

Remark (Structural View). Every λ-term has a tree structure:

- variables are leaves,
- abstractions $\lambda x. M$ are unary nodes,
- applications MN are binary nodes.

Thus Λ is a freely generated syntactic algebra.

Bibliography